Riya Kadakia

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Education

Carnegie Mellon University, Pittsburgh, PA
B.S. Robotics and Information Systems (GPA: 4.0/4.0, Dean's List), Class of 2028
Coursework: Intelligent Robot Systems, Robot Kinematics & Dynamics, Feedback Control Systems, Robot Building Practices

- *Available May - August 2027*

Research

Robotic Caregiving and Human Interaction Lab, Carnegie Mellon University - *Research Assistant* Mar 2025 – Aug 2025

- Co-author for RaC: Robot Learning for Long-Horizon Tasks by Scaling Recovery and Correction (accepted IEEE T-RO)
- Built the VR teleoperation and multi-camera capture system for collecting bimanual, contact-rich, long-horizon demonstrations.
- Human-in-the-loop recovery framework reached 78% success on real-world bimanual tasks with 10× data efficiency.

Momentum Robotics Lab, Carnegie Mellon University - *Research Assistant* Feb 2026 – Present

- Built a perception pipeline that uses vision-language models to detect and localize articulation points (joints) on mechanical objects; improved accuracy with Segment Anything pre-processing and grid-based coordinate refinement.
- Integrated the GR00T N1.6 vision-language-action model into the Newton physics simulator, mapping camera input and natural-language instructions to end-effector trajectories via a diffusion-transformer policy.
- Ran manipulation experiments on xArm7 and bimanual Franka FR3 platforms; implemented inverse kinematics, gripper control, and force-sensing feedback across varied objects.

Robotics Projects

Mobile Robot State Estimation, Mapping & Localization — *TurtleBot3, ROS 2, C++, Python* Fall 2026

- Fused gyro and wheel-encoder yaw via a complementary filter; tuned α over 20 trials, cutting heading error from 6.2° to 2.9°.
- Implemented point-to-point (SVD) and point-to-plane (KD-tree, normals) ICP to align LiDAR scans and reduce odometry drift.
- Built a log-odds occupancy-grid mapper; re-localized from a random pose with Monte Carlo localization in Nav2 and RViz.
- Debugged distributed ROS 2 issues including chrony clock sync for TF lookups and DDS domain/discovery configuration.

Autonomous Sorting Trash Bot — *CAD, 3D Printing, CNC, Laser Cutting* Fall 2026

- Created a robot from scratch, designing the end-effector and lift mechanism from cardboard prototype through CAD to fabricated parts, picking up and sorting paper, a plastic bottle, and a beanbag into separate bins.

Adaptive Gain-Scheduled PID for a Delivery Quadcopter — *MATLAB, Simulink* Spring 2026

- Replicated a published quadcopter model with 4 PID loops (altitude, roll, pitch, yaw) and validated open-loop hover dynamics.
- Showed fixed-gain PID failing under a 10× payload jump due to integrator windup (∼115% overshoot); designed mass-indexed gain scheduling that cut overshoot to ∼28% across 0.5–5 kg.

Franka Arm: Trajectory Generation to Light Painting — *Franka Panda, Python* Fall 2025

- Co-developed a trapezoidal-velocity joint-space trajectory generator verified with forward kinematics.
- Added a standby/recovery mode (continue, retry, safe exit) for repeatable, safe hardware operation.
- Capstone: planned Cartesian arc trajectories to light-paint a three-semicircle ocean wave with color-coded segments.

Experience

IBM — *Software Developer Intern* May 2026 – Aug 2026

- Built a lightweight gateway service routing between 10 models on a Spark/OpenShift cluster through a single endpoint, reducing model onboarding time by 80% and supporting scalable LLM integration.
- Developed a KVP extraction subpage for the Watson Document Understanding Demo site, integrating a semantic KVP extraction API with 8 built-in schemas and unlimited custom schema support for structured JSON output rendering.

Carnegie Mellon Racing (Driverless) — *Software Architecture Project Lead* Aug 2025 – May 2026

- Architect ROS 2 and OpenCV perception stack for real-time LiDAR-camera fusion, cutting manual calibration effort by 90%
- Built an automated extrinsic-calibration pipeline pairing 3D LiDAR points with 2D camera detections for the autonomous car.
- Wrote ROS 2 nodes that detect AprilTags, localize the board center, and convert LiDAR point clouds to NumPy for processing.
- Implemented a RANSAC-based transform solver and validation suite to keep multi-sensor alignment robust to outliers.

Technical Skills

- **Languages:** C++, Python, C, MATLAB/Simulink, SQL, Java
- **Perception & Estimation:** LiDAR–camera calibration, AprilTags, RANSAC, ICP, occupancy-grid mapping, particle filters, complementary filtering, VLMs, Segment Anything
- **Robotics Software:** ROS 2, Git, Docker, Gazebo, Nav2, tf2, RViz, MuJoCo, Newton, NeRF, Gaussian Splatting
- **Kinematics & Control:** Forward/inverse kinematics, trajectory generation, PID, gain scheduling, force feedback
- **Hardware:** Franka FR3/Panda, xArm7, TurtleBot3, Raspberry Pi, VR teleoperation, CAD, 3D printing, CNC, laser cutting